



The Adoption and Use of Artificial Intelligence and Machine Learning in Clinical Development

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Abstract

Background The use of artificial intelligence (AI) and machine learning (ML) in drug discovery has been well documented, but measures of levels of adoption, investments, and efficiencies gained from its use in clinical development have not yet been developed, captured or published. AI/ML use in clinical development is expected to increase, but its impact has not yet been systematically measured until now.

Methods The Tufts Center for the Study of Drug Development conducted a global online survey among pharmaceutical and biotechnology companies, contract research organizations (CROs), and data and technology vendors servicing drug developers. The survey gathered 302 responses assessing levels of AI/ML implementation across 36 distinct clinical trial planning and design, trial execution, and regulatory submission activities. The survey collected data on US dollar investment, time savings, and challenges and opportunities of AI/ML use in clinical development.

Results Approximately one-third of the sample (36.9%) was not yet using or implementing AI/ML across 36 design and planning, execution, and regulatory submission activities; another 30.3% was beginning their AI/ML implementation (or piloting), 22.1% was partially implementing (or moving beyond pilots), and on average only 10.7% had fully implemented AI/ML (i.e., uses AI in most trials employing a repeatable process).

Keywords Artificial intelligence · Machine learning · Generative AI · Quality by design · Clinical trial design · Innovation adoption · Adoption maturity · Protocol complexity · Protocol design · Clinical trials · Research and development · Clinical development · Real-world data

Introduction

The United States Food and Drug Administration (FDA) has outlined broad usage of artificial intelligence (AI) in the drug development process including multiple activities within clinical research such as patient recruitment, retention, and adherence; site selection; clinical trial data collection, management, and analysis; and endpoint assessment [1]. Recent FDA draft guidance defines AI as “a machine-based system that can, for a given set of human defined objectives, make predictions, recommendations, or decisions influencing real or virtual environments.” (p. 2) [2]. In addition, “Machine learning (ML) refers to a set of techniques that can be used to train AI algorithms to improve performance at a task based on data.” (p. 2) [2]. These definitions, applied broadly across drug development, will be used within the context of clinical development in this paper.

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The research on AI/ML usage within clinical research includes topics as diverse as optimization of trial eligibility and recruitment strategies, monitoring endpoints through wearables and surrogate endpoints, and improving operational efficiency. Studies have examined the use of AI in analyzing clinical trial eligibility criteria through electronic medical records (EMRs) and patient eligibility criteria databases, enhancing diversity of patient populations, and reducing sample sizes through implementation of adaptive trials [3–6]. Similarly, other applications have combined real-world data (RWD) and electronic health record (EHR) data to broaden eligibility criteria and potentially include patients that could benefit from a treatment [7]. In addition, AI is used for patient adherence control and endpoint detection and retention [5, 8]. AI is also used to analyze structured and unstructured data—such as clinical notes or medical records through natural language processing (NLP) [1, 5, 9]. AI also extracts real-world data from claims data, biobanks, patient registries, and imaging and digital pathology.⁵

An examination of 51 published articles (published between 2004 and 2024) found oncology was the most represented therapeutic area [10]. Improved efficiency and cost savings derived from using AI for patient recruitment and retention were reported in the literature. However, some challenges were also noted, including in the quality of the training data, concerns about data security and privacy, and selection bias [10].

An analysis of 48 publications on the use of AI and ML in clinical trials from the past five years in the U.S. and Europe revealed that oncology was the most prevalent therapeutic area for the use of AI [11]. A key finding of that analysis was that AI was applied more commonly in patient recruitment than in preclinical research, trial design, trial conduct, and analysis activities.

Implementation of AI within clinical trials also provides insights into country and site selection by predicting site recruitment and enrollment rates. The use of AI to forecast recruitment of patients has shown to increase precision in patient selection, and an increase in enrollment rates [7, 12–15]. In addition, models can allow trial planners to explore drivers of enrollment and analyze how timelines can potentially be accelerated by altering those factors.

AI is also implemented in patient monitoring to improve patient safety, and to reduce the risk of adverse events (AEs). AI algorithms can predict risks and AEs using in vivo and in vitro studies and can help detect adverse events during clinical trials and during post-marketing surveillance [16]. ML models have also been developed and used to detect adverse events that have been under-reported [17]. Other applications of AI in clinical trials involve digital health technologies; for example, deep learning and other ML models are used to interpret high-dimensional data from

patient wearables and video monitoring to predict various clinical endpoints [18, 19].

The use of AI in oncology trial enrollment has been compared to the results achieved through typical recruitment approaches [20]. Two completed breast and lung cancer studies, and one lung cancer study that failed to enroll, were examined. Use of AI would have resulted in a 24–50% increase in patients recruited compared to standard recruitment methods, by appropriately identifying potentially eligible patients in the enrolling studies. And it identified no suitable patients for a trial that was unable to enroll patients. The study also demonstrated the use of AI would have resulted in a significant reduction in the expected duration between patient eligibility and screening.

There are numerous challenges with incorporating AI into clinical trials, such as data quality and completeness [11], including Electronic Medical Record (EMR) data harmonization and interoperability problems. Lack of guidelines on integrating AI into trials are additional hurdles to AI utilization [21].

Other barriers to implementation of AI are data privacy, integrity and security [4]. Both the US Health Insurance Portability and Accountability Act (HIPAA) and the EU General Data Protection Regulation (GDPR) limit data use, especially with third parties, and can create legal and regulatory challenges [5, 22, 23]. Additionally, there are significant risks to patient privacy with accessing large amounts of data and involvement of third-party vendors.

Another challenge for the use of AI in clinical development is its explainability and ability to demonstrate value not only to those within the pharmaceutical industry but also to regulatory agencies [5]. Since some AI technologies are still in the early stages of adoption in clinical development, maturity in terms of adoption and usage has not yet been fully realized. Multi-disciplinary and multi-stakeholder efforts to gather insights into adoption of AI and ML, along with the challenges experienced, have led to key learnings that may advance industry usage. Among these are (1) the ability to provide use cases to increase industry confidence, (2) a regulatory framework to deal with ethical issues, and (3) continued collaboration among stakeholders including pharmaceutical, biotechnology and technology companies, regulators, academic institutions, and patients [24]. In addition to reviewing specific topics and uses of AI/ML, the FDA discussion paper calls attention to the need to engage diverse stakeholders in advancing AI [1], and the FDA draft guidance recommends early engagement with the FDA on AI models [2].

For all the efforts to characterize AI/ML use in clinical development, no study, to our knowledge, examines the use of AI/ML throughout clinical development and across a broad range of activities. To fill the gap, this study

examined adoption and usage of AI/ML across an expansive range of clinical development activities within a large sample of pharmaceutical and biotechnology companies, contract research organizations (CROs), and data and technology vendors servicing drug developers. We cataloged the various applications of AI/ML and ascertained stakeholders' use of various AI/ML approaches applied in trial design, planning, execution, and regulatory submission. We also assessed the resources required to implement AI/ML, the perceived impact on drug development timelines, and the general uptake of AI/ML use.

Methods

Study Design

The study used a custom questionnaire. The questionnaire was developed by the Tufts Center for the Study of Drug Development (Tufts CSDD) research team following the American Association for Public Opinion Research (AAPOR) [25] guideline, with input from the Drug Information Association (DIA) and a working group comprising fourteen pharmaceutical companies and two contract research organizations (CROs). The Institutional Review Board at Tufts University approved the study. Participants provided their electronic informed consent at the beginning of the questionnaire.

The questionnaire was designed and developed to be answered by individuals working at biotechnology companies, pharmaceutical companies, and contract research organizations and other vendors with familiarity and experience with AI/ML within the drug development context, and it was validated in several ways. First, the research team performed a literature review on studies that described typical activities included in the clinical development process that have been or may be impacted by the use AI/ML, starting with design and planning through execution, to regulatory submission. Second, the questions were shared with the working group of individuals from pharmaceutical companies and CROs who provided additional feedback. The working group included experts in clinical development, clinical operations, data sciences, quality and safety, and regulatory affairs—who provided additional feedback. Third, the questionnaire was piloted among a group of industry experts to ensure relevance and appropriateness. It was then distributed globally by the DIA, Tufts CSDD and working group members using Qualtrics, an online survey platform, between April 2024 to July 2024.

Measures

The questionnaire included 34 closed-ended and Likert-scale questions about company and functional characteristics, levels of adoption of AI/ML, investments in AI/ML, and perceived effectiveness and time reductions derived from AI/ML use across 36 unique clinical trial activities, including 12 design and planning activities, 20 execution activities, and 4 regulatory submission activities (Tables 1 and 2).

Statistical Analysis

A total 302 survey responses from drug development professionals were included in the analysis. Data cleaning and analysis were conducted by the Tufts CSDD research team using R statistical software [27]. Descriptive statistics and statistical testing for relationships between variables of interest, such as annual trial volume and stage of AI/ML adoption, were conducted to identify and benchmark applications of AI/ML to areas within clinical operations and development including study design, site identification and patient recruitment, pharmacovigilance, quality assurance, and clinical monitoring. For analysis purposes, categories within the stage of AI/ML adoption subgroup were reclassified into four stages: No Implementation (which includes “Planning but not using”, “Interested but not planning or using”, and “No interest”); Beginning Implementation; Partial Implementation; and Full Implementation). Similarly, companies were classified by annual trial volume into three categories: Fewer than 25; 25 to 100; and more than 100 trials. Additionally, the therapeutic area subgroup category was classified into Oncology (those who reported having experience in Oncology) and Other (those without experience in Oncology, but with experience in other therapeutic areas).

Further analyses were conducted to assess time reductions resulting from investments in the use of AI/ML in a typical development program. Frequencies and proportions were calculated. Subgroup analysis included functional area, company type (pharma / biotech, CRO, or other vendor), world region, annual trial volume, and therapeutic areas of experience. Where comparisons across subgroups were feasible and appropriate, P-values were reported; $P < .05$ was considered significant.

Results

A total of 302 respondents from 34 countries with familiarity with AI/ML in clinical development completed the survey and were included in the analysis, representing an approximate response rate of 8.2%. Table 3 presents the

Table 1 List of clinical trial activities assessed by clinical trial implementation stage

	Clinical Trial Activity Assessed ^a	Clinical Trial Implementation Stage
1	Budgeting and contracting	Design & Planning
2	Consent form development	Design & Planning
3	Content development for educational materials	Design & Planning
4	Identification of diverse populations	Design & Planning
5	Institutional Review Board (IRB) / Ethical Review Committee (ERC) submissions	Design & Planning
6	Literature reviews	Design & Planning
7	Protocol design – Real World Data (RWD) to inform eligibility criteria	Design & Planning
8	Protocol design – Real World Data (RWD) to inform endpoints	Design & Planning
9	Protocol design – writing	Design & Planning
10	Protocol design – other	Design & Planning
11	RBM/RBQM ^b – risk and quality assessment planning	Design & Planning
12	Site selection (including country selection)	Design & Planning
13	Chatbot – for virtual patient engagement (including for consent process)	Execution
14	Chatbot – for virtual site engagement	Execution
15	Data analysis – “omics” data such as proteomic data, metabolomic data, etc.	Execution
16	Data analysis – genetic data	Execution
17	Data analysis – imaging data	Execution
18	Data analysis – related to effect estimation	Execution
19	Data analysis – related to quality control or data cleaning	Execution
20	Drug and/or product order authenticity	Execution
21	Endpoint analysis	Execution
22	Patient eligibility	Execution
23	Patient enrollment assessments (including diversity assessments)	Execution
24	Patient monitoring	Execution
25	Patient narratives	Execution
26	Patient recruitment	Execution
27	RBM/RBQM ^b – audits, audit reports, and CAPA ^c resolution	Execution
28	RBM/RBQM ^b – central monitoring	Execution
29	RBM/RBQM ^b – risk assessments / pharmacovigilance / safety	Execution
30	Shipping and manufacturing accountability	Execution
31	Site monitoring (i.e., study team monitoring)	Execution
32	Social listening for adverse event reporting	Execution
33	Clinical Study Report (CSR) writing	Regulatory Submission
34	ISO IDMP ^d component gathering	Regulatory Submission
35	Regulatory submissions	Regulatory Submission
36	Trial Master File (TMF) filing	Regulatory Submission

^aThis is the list of activities that may be supported by AI/ML throughout the clinical development process. The questionnaire inquired into levels of adoption and time savings achieved via implementation of AI/ML in each of the activities included in the table

^bRBM/RBQM means Risk-based Monitoring and Risk-based Quality Management

^cCAPA means Corrective and Preventive Actions

^dISO is the International Organization for Standardization and IDMP is Identification of Medicinal Products

distribution of participants by world region, company type, annual trial volume, functional area, and therapeutic area of experience. Most respondents resided in North America (51.4%) and Europe (22.4%). More than half of respondents (55.9%) represented biopharmaceutical companies while 20.2% represented a CRO and 23.9% represented a type of vendor—a medical device company, a technology vendor, or data vendor. The distribution of the sample by annual trial volume was 39.4% representing smaller companies conducting fewer than 25 trials annually, 23.0% representing

medium-sized companies conducting 25–100 trials annually, and 37.6% representing larger companies conducting more than 100 trials annually. The sample comprised 38.8% representing clinical operations or clinical development, and 37.8% representing data science, data management, or data quality. Nearly half of the sample (41.1%) reported experience in oncology. The rest of the sample reported having experience in at least one of the following therapeutic areas but no experience in oncology: urology, cardiovascular and circulatory diseases, immunology, neurology/

Table 2 Survey definitions

Term ^a	Definition
Artificial Intelligence (AI)	“AI can generally be described as a branch of computer science, statistics, and engineering that uses algorithms or models to perform tasks and exhibit behaviors such as learning, making decisions, and making predictions.” ¹
Machine Learning (ML)	“ML is considered a subset of AI that allows ML models to be developed by ML training algorithms through analysis of data, without models being explicitly programmed. Additionally, there are a variety of ML methods and different types of algorithms that may be utilized in a given context.” ¹ For this study, AI and ML are referenced together as AI/ML.
Generative Artificial Intelligence (GenAI)	“GenAI systems create new content – including text, image, audio, and video – based on their training data and in response to prompts.” [26]
Stage of AI/ML Implementation ^a	Definition
Fully Implemented	“A repeatable process is in place” for AI/ML use
Partially Implemented	“Moving beyond pilots” for AI/ML
Beginning Implementation	“Piloting” AI/ML usage
Planning but not using	“Planned use within the next 24 months”
Interested but not planning or using ^b	
No interest ^b	

^aDefinitions were presented to respondents at the beginning of the questionnaire to ensure familiarity with artificial intelligence and machine learning, and that questions were answered based on a harmonized understanding of the concepts

^bDefinition was not provided

neurodegenerative diseases, rare diseases and disorders, gene therapies, infectious diseases, hematology, inflammation, dermatology, vaccines, respiratory and pulmonary diseases, ophthalmology, endocrinology, gastroenterology, genetics, nephrology, psychiatry, internal medicine, rheumatology, musculoskeletal, sleep, fertility and reproductive health, obstetrics and gynecology, dental and oral health, nutrition, orthopedics, podiatry, plastic surgery.

Use of AI/ML in Clinical Development

Across 36 design and planning, execution, and regulatory submission activities, 36.9% of the sample was not using or implementing AI/ML, another 30.3% was beginning their AI/ML implementation (i.e., piloting), 22.1% was partially implementing (i.e., moving beyond pilots) and only 10.7% had fully implemented AI/ML (i.e., uses AI in most trials employing a repeatable process). In 2024, ‘beginning implementation’ of AI/ML was highest among regulatory submission activities (31.8%) and lowest among execution activities (29.0%); ‘partial implementation’ was highest among execution activities and lowest among regulatory submission activities; and full implementation was highest among execution activities and lowest among design and planning activities (see Table 1). Notable differences were observed in AI/ML implementation by Annual Trial Volume across the various stages of clinical development. Table 4 presents levels of AI/ML employment overall and by Annual Trial Volume.

Implementation of AI/ML varied across the 36 clinical development activities assessed (Table 5), as well as by

annual trial volume. The activities with the highest levels of partial (i.e., moving beyond pilots) or full implementation were ‘identification of diverse populations’ (51.6%), ‘data analysis – genetic data’ (50.0%), and ‘patient narratives’ (47.9%). Activities where absence of implementation was highest included ‘ISO IDMP component gathering’ (75.0%), ‘shipping and manufacturing accountability’ (71.5%), and ‘budgeting and contracting’ (70.3%).

The highest levels of full implementation were observed in ‘patient narratives’ and ‘data analysis – genetic data’, each at 20.8% and 19.2% respectively.

Table 6 ranks the top six clinical trial activities by highest AI/ML implementation levels. It also ranks activities by highest levels of absence of AI/ML implementation.

Investment and Time Reductions Associated with the Use of AI/ML

In addition to collecting data on levels of AI/ML implementation, the study gathered information on the US dollar investment required to deploy tools to support AI/ML use across 14 of 36 clinical trial activities included in the assessment. Table 7 depicts the mean investment allocation toward use of AI/ML in each of the 14 activities, across design and planning, execution, and regulatory submission. The highest mean levels of investment were observed in ‘data analysis – related to quality control or data cleaning’ at US \$3.2 million and in ‘trial master file (TMF) filing’ at US \$2.2 million, followed by ‘site selection (including country selection)’ at US \$1.9 million.

Table 3 Respondent characteristics

	<i>n</i>	%
World Region		
North America	149	51.4%
Europe	76	26.2%
ROW ^a	65	22.4%
Company Type		
Pharma / Biotech	166	55.9%
CRO	60	20.2%
Other vendors ^b	71	23.9%
Annual Trial Volume		
Fewer than 25	111	39.4%
25 to 100	65	23.0%
More than 100	106	37.6%
Functional Area		
Clinical Operations/Clinical Development	111	38.8%
Data Science/Data Management/Data Quality	108	37.8%
Other ^d	67	23.4%
Therapeutic Areas of Experience		
Oncology	124	41.1%
Other ^e	178	58.9%

^aIncludes India, China, Japan, Brazil, Australia, Israel, Afghanistan, Ethiopia, Kazakhstan, Nigeria, Peru, Philippines, Saudi Arabia, South Africa

^bIncludes medical device or diagnostics companies, data vendors, and technology vendors

^cIncludes biostatistics/biometrics, computer science/information technology, data engineering, data management, data science/advanced analytics, health economics & outcomes research (HEOR), pharmacovigilance/safety/risk management, real world evidence (RWE), quality assurance - GXP

^dIncludes marketing/sales/business development, outsourcing/vendor management/sourcing/procurement, regulatory compliance/legal, and “other” functions (unspecified by respondents)

^eIncludes urology, cardiovascular and circulatory diseases, immunology, neurology/neurodegenerative diseases, rare diseases and disorders, gene therapies, infectious diseases, hematology, inflammation, dermatology, vaccines, pulmonary and respiratory diseases, ophthalmology, endocrinology, gastroenterology, genetics, nephrology, psychiatry, internal medicine, rheumatology, musculoskeletal, sleep, fertility and reproductive health, obstetrics and gynecology, dental and oral health, nutrition, orthopedics, podiatry, plastic surgery

Similarly, data on time reductions achieved with the use of AI/ML was obtained for 26 out of the 36 activities included in the assessment (Table 8). On average, companies derived 18% time reductions when AI/ML was used to support clinical trial activities. Reported time reductions differed throughout the clinical trial continuum. Mean time reductions were highest in regulatory submission activities (-22%), followed by execution activities (-20%) and design & planning activities (-13%). Overall, of all activities assessed, the highest time gains were observed in ‘patient monitoring’ (-75%), ‘identification of diverse populations’ (-68%), and ‘regulatory submissions’ (-63%). Increased timelines were only observed in one activity when supported

by AI/ML use: ‘site selection (including country selection)’ with a 13% mean timeline increase.

Perceptions on AI/ML Use and Implementation

The study collected data on respondent perceptions related to familiarity, organizational trust in AI/ML recommendations, and confidence in the quality and cleanliness of the data used to train machine learning models. Information on perceived challenges to AI/ML implementation and opportunities to improve trust was also collected.

Table 9 presents respondent perceptions on familiarity with AI/ML and organizational trust/confidence, with a focus on percentage levels among those most likely (very or somewhat/moderately) to be familiar with AI and GenAI, as well as on those most likely (very or somewhat) to believe that their organizations are confident and trusting of AI/ML. Overall, 90% of respondents were familiar with AI and 78% were familiar with GenAI. Lower levels of confidence and trust were observed at the organizational level; 68% believe their organization is confident that the underlying data used to train AI/ML is accurate or clean, while 61% trust the recommendations or outcomes of the AI/ML tools.

The top reported challenge to AI/ML implementation was ‘data quality’ (41%), including data locks, low quality and/or quantity of data available for AI/ML. The next most reported challenge was lack of ‘trust in AI/ML-generated output’ (37%), including data-driven or algorithmic biases, and lack of trust in AI-generated results or recommendations. Finally, the most reported methods for increasing trust in AI/ML included ‘more experiences with successful implementations/use cases’ (48%), followed by a clearly articulated return on investment (ROI) (Table 10).

Discussion

This study demonstrates the breadth of activities that professionals in clinical development consider under the umbrella of AI/ML. A key finding is that use of AI/ML across activities involving trial design and planning, execution and regulatory submissions is predominantly in the beginning stages of implementation. These results are consistent with findings in the published literature that AI use within clinical development has not yet reached maturity [5, 24]. In addition, over a third of respondents surveyed were not using AI/ML, based on the categories of ‘no interest’ and ‘interested but not planning or using,’ although some are ‘planning but not using’.

Table 4 Rates of implementation by clinical development stage and annual trial volume

Annual Trial Volume	Design & Planning		Execution		Regulatory Submission		Aggregate	
	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%
Overall								
<i>n</i> = 1080 ^a								
No implementation ^b	156	37.8%	193	35.9%	49	38.0%	398	36.9%
Beginning implementation	130	31.5%	156	29.0%	41	31.8%	327	30.3%
Partial implementation	90	21.8%	123	22.9%	26	20.2%	239	22.1%
Full implementation	37	9.0%	66	12.3%	13	10.1%	116	10.7%
Fewer than 25 trials								
<i>n</i> = 306 ^a								
No implementation ^b	59	49.2%	70	46.1%	18	52.9%	147	48.0%
Beginning implementation	38	31.7%	34	22.4%	9	26.5%	81	26.5%
Partial implementation	12	10.0%	19	12.5%	4	11.8%	35	11.4%
Full implementation	11	9.2%	29	19.1%	3	8.8%	43	14.1%
25 to 100 trials								
<i>n</i> = 190 ^a								
No implementation ^b	30	42.9%	24	27.0%	15	48.4%	69	36.3%
Beginning implementation	19	27.1%	27	30.3%	4	12.9%	50	26.3%
Partial implementation	16	22.9%	28	31.5%	8	25.8%	52	27.4%
Full implementation	5	7.1%	10	11.2%	4	12.9%	19	10.0%
More than 100 trials								
<i>n</i> = 495 ^a								
No implementation ^b	60	31.7%	86	34.0%	13	24.5%	159	32.1%
Beginning implementation	61	32.3%	79	31.2%	22	41.5%	162	32.7%
Partial implementation	48	25.4%	63	24.9%	12	22.6%	123	24.8%
Full implementation	20	10.6%	25	9.9%	6	11.3%	51	10.3%

^aAI/ML reported implementation level observations across 36 activities^bIncludes observations reporting any of the following: “No interest”, “Interested but not planning or using”, “Planning but not using (Planned use within the next 24 months)”

There is substantial variation in stage of implementation across activity types and organizational characteristics. For example, respondents whose companies were conducting more than 100 trials annually were more likely to report full or partial implementation of AI/ML solutions in trial design and planning activities than those with smaller trial volumes.

Some clinical development activities have moved beyond beginning implementation to partial and full implementation, many of which are focused on data analysis and data quality. The areas with the highest levels of implementation include identification of diverse populations; data analysis (genetic data); patient narratives; risk-based monitoring and risk-based quality management (RBM/RBQM); and data analysis for imaging and omics data (i.e., genomic, proteomic, metabolic and interatomic data). Data analysis related to quality control or data cleaning was reported as having the highest level of financial investment. Implementation of AI/ML for RBM/RBQM has been noted as an effective alternative to traditional monitoring due to its ability to increase the quality of the data generated by trial sites [28]. Some of the activities with the highest frequency of full implementation have the most well-developed methodology

for application to drug development. For example, 20.8% of respondents reported full implementation of ML for patient narratives. NLP methods and software have been well developed for common use cases such as detecting drug exposures and adverse events [29] in spontaneous reports and online forums [30].

The most significant findings regarding time reductions reported for activities using AI/ML are in patient monitoring; identification of diverse populations; regulatory submissions; patient enrollment assessments including diversity assessments; data analysis (omics data); and protocol design focused on RWD to inform endpoints. The latter activity was also among the top areas of financial investment by companies. A notable result is that the use of AI/ML has led to significant time reductions in patient monitoring by facilitating more effective examination, as well as earlier detection of adverse events and safety concerns [18, 19, 31].

The ability to decrease the time spent on ensuring diverse representation of patients in clinical trials and establishing diversity enrollment assessments can be enhanced through AI/ML. Diverse clinical trials ensure that the study population accurately reflects the real-world population and that

Table 5 Levels of implementation of AI/ML by clinical development activity

	No implementation ^b	Beginning implementation	Partial implementation	Full implementation
<i>n</i> = 1080 ^a				
Design and Planning				
Budgeting and contracting	70.3%	18.5%	3.7%	7.4%
Institutional Review Board (IRB) / Ethics Review Committee (ERC) submissions	69.3%	23.1%	0.0%	7.7%
Consent form development	54.5%	36.4%	3.0%	6.1%
Protocol design – writing	43.2%	39.2%	13.7%	3.9%
RBM/RBQM ^c – risk and quality assessment planning	40.0%	20.0%	33.3%	6.7%
Protocol design – other	36.9%	36.8%	15.8%	10.5%
Protocol design – Real World Data (RWD) to inform endpoints	32.3%	29.0%	29.0%	9.7%
Content development for educational materials	32.1%	28.3%	28.3%	11.3%
Protocol design – Real World Data (RWD) to inform eligibility criteria	31.0%	34.5%	27.6%	6.9%
Site selection (including country selection)	29.7%	32.4%	21.6%	16.2%
Literature reviews	28.5%	35.1%	23.0%	13.5%
Identification of diverse populations	22.6%	25.8%	51.6%	0.0%
Execution				
Shipping and manufacturing accountability	71.5%	14.3%	0.0%	14.3%
Drug and/or product order authenticity	66.6%	33.3%	0.0%	0.0%
RBM/RBQM ^c – Audits, audit reports, and CAPA ^d Resolution	50.0%	40.0%	10.0%	0.0%
Chatbot – for virtual Patient engagement (including for consent process)	48.4%	22.6%	19.4%	9.7%
RBM/RBQM ^c – risk assessments/ pharmacovigilance/ safety	44.8%	27.6%	17.2%	10.3%
Site monitoring (i.e., study team monitoring)	43.8%	18.8%	31.2%	6.2%
Endpoint analysis	40.6%	37.5%	9.4%	12.5%
RBM/RBQM ^c – Central Monitoring	37.9%	17.2%	27.6%	17.2%
Chatbot – for virtual Site engagement	37.6%	25.0%	29.2%	8.3%
Data analysis – related to quality control or data cleaning	37.5%	37.5%	18.8%	6.2%
Patient enrollment assessments (including diversity assessments)	37.5%	25.0%	31.2%	6.2%
Patient monitoring	37.0%	29.6%	25.9%	7.4%
Patient recruitment	35.7%	28.6%	21.4%	14.3%
Social listening for adverse event reporting	34.7%	30.4%	17.4%	17.4%
Data analysis – genetic data	30.7%	19.2%	30.8%	19.2%
Patient eligibility	30.7%	34.6%	23.1%	11.5%
Data analysis – imaging data	29.4%	26.5%	26.5%	17.6%
Data analysis – related to effect estimation	27.3%	42.4%	21.2%	9.1%
Patient narratives	27.1%	25.0%	27.1%	20.8%
Data analysis – “omics” data such as proteomic data, metabolomic data, etc.	21.8%	34.4%	28.1%	15.6%
Regulatory Submission				
ISO IDMP ^e component gathering	75.0%	12.5%	0.0%	12.5%
Regulatory submissions	54.2%	20.0%	17.1%	8.6%
Trial Master File (TMF) filing	30.8%	30.8%	26.9%	11.5%
Clinical Study Report (CSR) writing	26.7%	41.7%	21.7%	10.0%

^aAI/ML reported implementation level observations across 36 activities (*n* = 1080)^bIncludes observations reporting any of the following: “No interest”, “Interested but not planning or using”, “Planning but not using (Planned use within the next 24 months)”^cRBM/RBQM means Risk-based Monitoring and Risk-based Quality Management^dCAPA means Corrective and Preventive Actions^eISO is the International Organization for Standardization and IDMP is Identification of Medicinal Products

Table 6 Ranking of clinical trial activities by highest and lowest levels of implementation

Rank	Clinical Trial Activities	Partial / Full Implementation (Percent of Total)
1	Identification of diverse populations	51.6%
2	Data analysis – genetic data	50.0%
3	Patient narratives	47.9%
4	RBM/RBQM ^a – Central Monitoring	44.8%
5	Data analysis – imaging data	44.1%
6	Data analysis – “omics” data such as proteomic data, metabolomic data, etc.	43.7%
Rank	Clinical Trial Activities	No Implementation ^b (Percent of Total)
1	ISO IDMP ^c component gathering	75.0%
2	Shipping and manufacturing accountability	71.5%
3	Budgeting and contracting	70.3%
4	Institutional Review Board (IRB) / Ethics Review Committee (ERC) submissions	69.3%
5	Drug and/or product order authenticity	66.6%
6	Consent form development	54.5%

^aRBM/RBQM means Risk-based Monitoring and Risk-based Quality Management

^bIncludes observations reporting any of the following: “No interest”, “Interested but not planning or using”, “Planning but not using (Planned use within the next 24 months)”

^cISO is the International Organization for Standardization and IDMP is Identification of Medicinal Products

variability in response due to genetic, lifestyle and environmental factors can be understood during the clinical development process. The Food and Drug Omnibus Reform Act (FDORA) of 2022, passed by Congress at the end of 2022, mandates that sponsors submit Diversity Action Plans (DAPs) for Phase III and pivotal studies [32]. This legislation follows FDA draft guidance previously released which provided recommendations for sponsors on developing racial and ethnic diversity programs for clinical trials to ensure adequate numbers of participants from underrepresented populations [33].

The greatest challenges reported by respondents in the survey were data quality challenges; trust in AI/ML general output; intellectual property and other legal concerns related to data sharing and issues related to the governance of ethical and privacy concerns. Data quality and integrity of data has been documented as a key challenge in literature as well as having insufficient data for AI models such as those used in ML [5, 11]. In addition, trust emerged as a key concern across organizations including data-driven or algorithmic bias and lack of trust in AI generated results, forecasts and recommendations. Querying respondents about approaches to potentially enhance trust highlighted several key factors: greater awareness and experience with successful implementations and use cases; the establishment of a clearly defined return on investment (ROI); and a focus on ethics, transparency, and explainability. Publications have investigated the barriers of AI use in its ability to explain and demonstrate value, especially for regulatory agencies and various stakeholders conducting clinical trials, noting that continued collaboration is critical [5, 24]. The involvement of third-party vendors in the clinical development process

Table 7 Investments in AI/ML use across clinical development stages

	<i>n</i>	Mean US Dollars Invested ^a	Coefficient of Variation	Range
Design & Planning				
Site selection (including country selection)	7	\$1,935,714	1.1	\$250,000–\$5,000,000
Protocol design – Real World Data (RWD) to inform eligibility criteria	5	\$1,300,000	0.9	\$200,000–\$3,000,000
Protocol design – Real World Data (RWD) to inform endpoints	4	\$1,100,000	1.0	\$100,000–\$2,000,000
Protocol design – writing	5	\$860,000	1.0	\$100,000–\$2,000,000
Identification of diverse populations	4	\$600,000	0.8	\$100,000–\$1,000,000
Content development for educational materials	4	\$225,000	1.0	\$50,000–\$500,000
Literature reviews	8	\$174,425	1.2	\$400–\$500,000
Execution				
Data analysis – related to quality control or data cleaning	8	\$3,231,300	2.1	\$400–\$20,000,000
Patient narratives	4	\$843,750	1.7	\$75,000–\$3,000,000
Patient enrollment assessments (including diversity assessments)	6	\$733,333	0.9	\$100,000–\$2,000,000
Patient recruitment	4	\$577,500	1.1	\$10,000–\$1,500,000
Data analysis – related to effect estimation	5	\$292,000	0.7	\$10,000–\$500,000
Regulatory Submission				
Trial Master File (TMF) filing	4	\$2,200,000	1.5	\$300,000–\$7,000,000
Clinical Study Report (CSR) writing	12	\$559,167	1.0	\$10,000–\$1,500,000

^aSample was normalized to only include activities with at least 4 observations within two standard deviations; 14/36 activities met these criteria

Table 8 Time reductions from AI/ML use by clinical trial activity

	<i>n</i>	Mean Time Reduction ^a	Coefficient of Variation	Range
Average Time Reduction	340	-18.0%	391.4	-500%-80%
Design & Planning				
Identification of diverse populations	9	-68.0%	241.4	-500%-30%
Protocol design – Real World Data (RWD) to inform endpoints	7	-26.0%	119.3	-70%-8%
Consent form development	10	-12.0%	168.4	-50%-10%
Protocol design – writing	14	-10.0%	339.0	-70%-50%
Literature reviews	30	-9.0%	350.1	-75%-80%
Content development for educational materials	17	-5.0%	720.6	-60%-65%
Budgeting and contracting	8	-3.0%	543.7	-30%-25%
Protocol design – Real World Data (RWD) to inform eligibility criteria	7	-1.0%	3053.0	-50%-50%
Site selection (including country selection)	9	13.0%	214.7	-20%-50%
Average - Design & Planning	111	-13.0%	449.9	-500%-80%
Execution				
Patient monitoring	7	-75.0%	251.5	-500%-40%
Patient enrollment assessments (including diversity assessments)	10	-45.0%	360.1	-500%-50%
Data analysis – “omics” data such as proteomic data, metabolomic data, etc.	9	-29.0%	66.1	-50%-10%
Patient narratives	21	-24.0%	127.7	-80%-40%
Data analysis – imaging data	12	-20.0%	164.6	-85%-20%
Data analysis – related to effect estimation	11	-18.0%	187.3	-90%-20%
Data analysis – related to quality control or data cleaning	21	-18.0%	185.8	-75%-50%
RBM/RBQM ^b – central monitoring	11	-15.0%	202.4	-60%-50%
Social listening for adverse event reporting	7	-11.0%	302.5	-70%-30%
Chatbot – for virtual Site engagement	7	-6.0%	449.5	-50%-30%
Endpoint analysis	10	-4.0%	608.0	-50%-30%
Patient recruitment	9	-3.0%	589.3	-25%-30%
RBM/RBQM ^b – risk assessments / Pharmacovigilance / Safety	11	-2.0%	1236.5	-25%-50%
Average - Execution	146	-20.0%	348.9	-500%-50%
Regulatory Submission				
Regulatory submissions	9	-63.0%	267.4	-500%-50%
ISO IDMP ^c component gathering	4	-13.0%	214.9	-50%-9%
Clinical Study Report (CSR) writing	26	-8.0%	407.4	-90%-60%
Trial Master File (TMF) filing	9	-3.0%	601.0	-20%-30%
Average - Regulatory Submission	48	-22.0%	438.2	-500%-60%

^aAI/ML reported time reduction observations across 26/36 activities

^bRBM/RBQM means Risk-based Monitoring and Risk-based Quality Management

^cISO is the International Organization for Standardization and IDMP is Identification of Medicinal Products

as well global regulatory concerns and differing regional policies also contribute to additional hurdles with AI use.

Trust was also examined in relation to respondents' familiarity with AI and GenAI. Significant differences were found among those familiar with AI in two areas of respondent perceptions about their organizations: confidence that the underlying data used to train AI/ML models is clean and accurate; and trust in the recommendations of the AI/ML solution over a human's. These differences were found across company types, respondent regions, annual trial volume of respondent organizations and functional areas. Research has indicated that for users to trust AI they need to develop knowledge and understanding about how these systems make decisions [34]. Respondents who have already

gained an understanding of AI/ML would likely be more confident and trusting concerning the use and application of these models. Establishing clear ROI and demonstrating successful use-cases that impact drug development processes through measures including quality and time can further build trust in AI/ML solutions. The results may suggest that upskilling or reskilling staff on AI/ML methods and systems could potentially increase trust and confidence in AI/ML methods. Research also indicates that AI can create new positions, competencies, and skill sets across health-care such as data governance experts and chief AI officers [35]. These roles can facilitate integration and effective usage of AI/ML within organizations.

Table 9 Perspectives on familiarity and trust

Very or Somewhat/Moderately	Familiar with the term AI ^a		Familiar with the term GenAI ^a		Believes organization is confident that the underlying data used to train AI/ML is accurate or clean ^b		Believes organization trusts recommendations or outcomes of the AI/ML solution ^c	
	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%
Overall	272	90.4%	231	77.5%	129	67.5%	118	60.8%
Region								
Europe	72	96.0%	62	82.7%	35	76.1%	29	63.0%
North America	143	96.0%	120	80.5%	63	70.8%	58	62.4%
ROW	47	72.3%	41	66.1%	27	57.4%	26	55.3%
Company Type								
Pharma / Biotech	155	93.4%	136	81.9%	69	63.9%	62	56.4%
CRO	47	79.7%	32	56.1%	28	73.7%	26	66.7%
Other Vendors	65	91.5%	59	84.3%	31	72.1%	30	69.8%
Annual Trial Volume								
Fewer than 25	102	91.9%	84	75.7%	42	60.0%	41	58.6%
25 to 100	58	89.2%	45	69.2%	33	82.5%	23	53.5%
More than 100	97	91.5%	87	83.7%	46	65.7%	48	68.6%
Functional Area								
Clinical Operations/Clinical Development	98	88.3%	85	77.3%	43	58.9%	44	58.7%
Data Science/Data Management/Data Quality	94	87.9%	81	77.1%	48	69.6%	40	58.8%
Other	66	98.5%	57	85.1%	30	73.2%	26	61.9%
Therapeutic Areas of Experience								
Oncology	113	91.9%	98	80.3%	55	70.5%	48	60.8%
Other	159	89.3%	133	75.6%	74	65.5%	70	60.9%

^aRespondents reporting they were “Very” and “Somewhat/Moderately” familiar

^bRespondents reporting their organizations are “Very” and “Somewhat” confident

^cRespondents reporting their organizations are “Very” and “Somewhat” trusting

The findings from the research align with what was anticipated, reinforcing ongoing themes in the discussion of AI application within drug development. For over a decade, the use of artificial intelligence in this field has been predicted to revolutionize pharmaceutical research, yet the anticipated breakthroughs have often fallen short. The introduction of large language models (LLMs) and generative AI has heightened expectations, but whether these technologies will truly deliver remains an open question [36]. The drug development enterprise is heavily regulated. Existing regulatory frameworks such as GDPR already pose significant challenges; the ambiguity of forthcoming AI regulations introduces additional barriers to the application of these technologies [37]. One lesson from the findings is the return on investment (RoI) that can be achieved with statistical models in pharmaceutical R&D [38]. These models are not only less computationally demanding but also offer transparency, a critical factor in navigating regulatory landscapes and making informed decisions. In drug development and clinical trials, transparent models can significantly enhance decision-making processes, ensuring that regulatory compliance is met more easily and effectively [39].

The study had some limitations. One was in terms of sampling. While the distribution aimed to reach individuals

through collaboration with a working group of sixteen biopharmaceutical companies and CROs, additional contacts were broadly and generally targeted across drug development to enhance response rates. However, this approach may have included individuals not familiar with AI in the outreach, resulting in fewer responses to the survey. The response rate was low but did cover a wide geographic area and company types and sizes. Another limitation is the response to the mix of questions focused on organizational perceptions and respondents’ individual expertise. Perceptions about respondent organizations may not have been accurately answered as some of the individual respondents may not have had comprehensive knowledge of their organization’s AI use, related time savings, or investments required. Other key questions, however, could have elicited more precise responses as they were structured to ask about respondent areas of expertise with AI/ML and provided responses based on these specific areas only.

Considering the rapidly evolving landscape of AI/ML, a similar study could be conducted again in 12 to 18 months to analyze changes in usage across clinical development activities. During that period, the findings from a survey, combined with additional use cases, could enable researchers to provide deeper insights into the development of AI/

Table 10 Challenges to implementation and methods to increase trust in AI/ML

Reported Challenges (<i>n</i> = 221) ^a	<i>n</i>	%
Data quality challenges (<i>such as data locks that prevent the use of data for machine learning models / low quality of data for machine learning / low quantity of data available for machine learning</i>)	91	41.2%
Trust in AI/ML generated output challenges (<i>such as data-driven or algorithmic bias / lack of trust in AI/ML-generated results, forecasts, and recommendations</i>)	82	37.1%
Intellectual Property (IP) and other legal concerns related to data sharing	65	29.4%
Issues related to the governance of ethical and privacy concerns (<i>such as conflicting federal/national/multilateral policies for organizations that work across countries, general lack of federal/national/multilateral policy, etc.</i>)	62	28.1%
Reported Methods to Increase Trust in AI/ML (<i>n</i> = 215) ^b	<i>n</i>	%
More experience with successful implementations/ use cases	104	48.4%
A clearly articulated Return on Investment (ROI)	76	35.4%
Ethics, Transparency, and Explainability	66	30.7%

^aRespondents selecting at least one challenge. Table shows options selected by at least 25% of respondents

^bRespondents selecting at least one method to increase trust. Table shows options selected by at least 25% of respondents

ML as well as reflect any outcomes from updated regulatory guidance for all stakeholders. Updated and new regulatory guidance could also be examined as new developments continue to be published and impact AI/ML adoption.

Conclusions

In this study, we found that overall pharmaceutical companies are primarily in the beginning stages of AI/ML implementation in clinical development. We also found that larger pharmaceutical companies—those with higher annual trial volume—are using AI/ML more vigorously across clinical trial activities. Overall, across all types of organizations, higher levels of partial or full AI/ML implementation were observed in the identification of diverse populations, genetic data analysis, capture and analysis of patient narratives, central monitoring as part of RBM/RBQM, analysis of imaging data, and analysis of “omics” data. AI/ML adoption strategies are crucial for advancing the drug development industry. Adoption relies on greater sharing of use cases, as well as of successes and failures among stakeholders engaged in the clinical development process, particularly given the low level of standardization regarding measurability of gains related to AI/ML implementation.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Supplemental Materials Questionnaire is available upon request.

Competing Interests The authors declare no competing interests.

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